

# Homework 10: Nonnegative Matrix Factorization

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## Problem 24: Uniqueness of NMF solutions

The factor matrices  $W$  and  $H$  in a non-negative matrix factorization  $V \approx WH$  are **not unique**. For any invertible  $K \times K$  matrix  $Q$  such that both  $Q$  and  $Q^{-1}$  have non-negative entries, the matrices

$$\tilde{W} = WQ \in \mathbb{R}_{\geq 0}^{M \times K}, \quad \tilde{H} = Q^{-1}H \in \mathbb{R}_{\geq 0}^{K \times N}$$

satisfy  $\tilde{W}\tilde{H} = WQQ^{-1}H = WH$  but in general  $\tilde{W} \neq W$  and  $\tilde{H} \neq H$ . Such matrices  $Q$  are the monomial matrices, i.e. products of a permutation matrix and a positive diagonal matrix.

**Concrete counter-example, diagonal scaling** Let  $D = \text{diag}(d_1, \dots, d_K)$  with  $d_k > 0$  and not all  $d_k = 1$ . Then

$$\tilde{W} := WD, \quad \tilde{H} := D^{-1}H$$

are non-negative because  $W, H$  are non-negative and  $D, D^{-1}$  are positive diagonal, differ from  $W$  and  $H$ , and

$$\tilde{W}\tilde{H} = (WD)(D^{-1}H) = W(DD^{-1})H = WH.$$

For example with  $K = 2$ , taking

$$W = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

gives  $\tilde{W} = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$  and  $\tilde{H} = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ , which are still non-negative and yield the same product  $WH = \tilde{W}\tilde{H} = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$ .

**Permutation ambiguity** Likewise, for any  $K \times K$  permutation matrix  $P$  which satisfies  $P^{-1} = P^T \geq 0$ , the pair  $\tilde{W} = WP$ ,  $\tilde{H} = P^TH$  reorders the components without changing the product. Combining the two yields the full monomial ambiguity  $\tilde{W} = WPD$ ,  $\tilde{H} = D^{-1}P^TH$ .

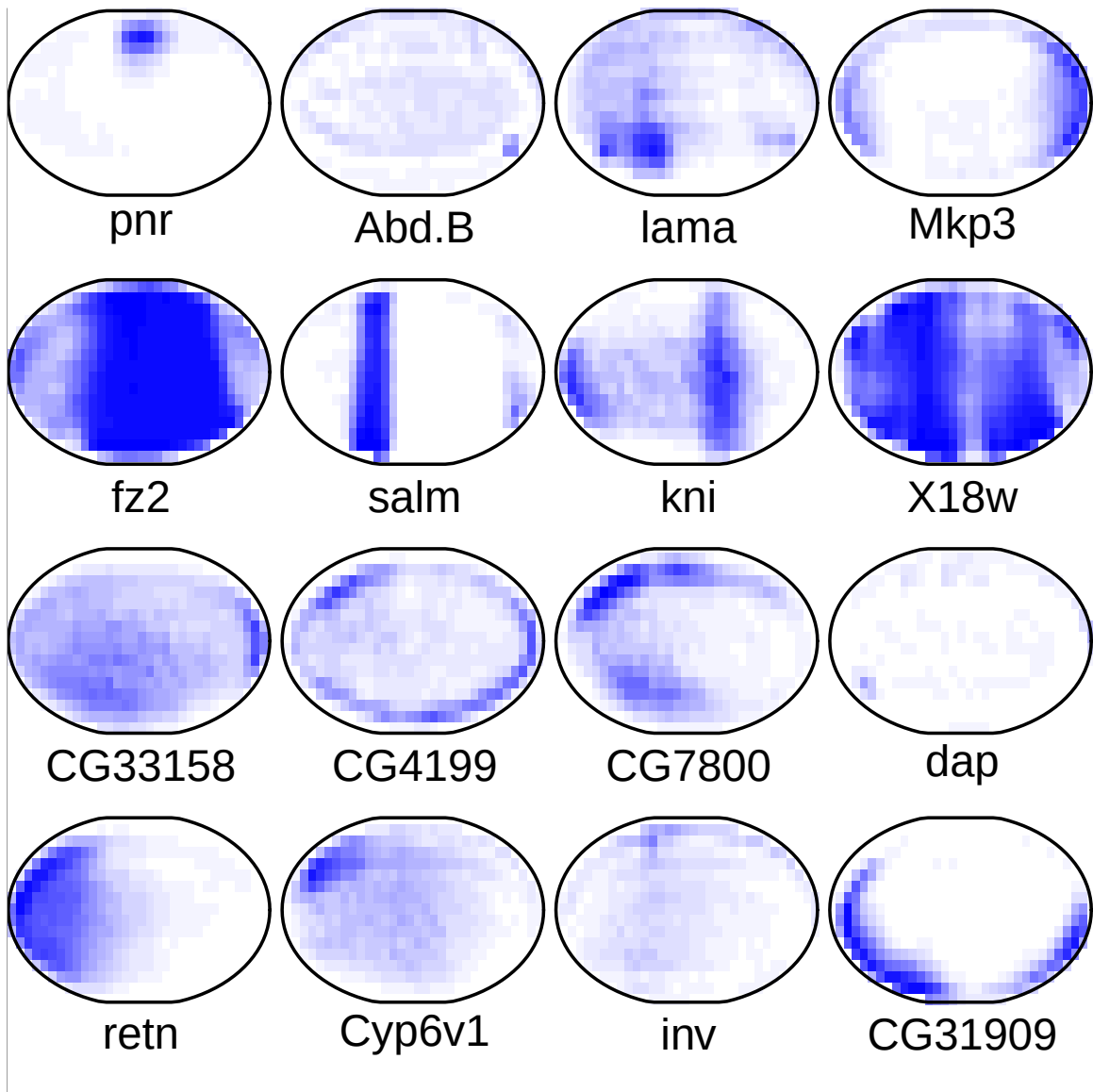
Hence NMF factors are unique only up to a permutation and rescaling of the components. Beyond these trivial ambiguities, additional non-uniqueness can occur for specific data, but the simple counter-examples above already suffice to disprove uniqueness in general.

## Problem 25: NMF of spatial gene expression patterns

```
suppressPackageStartupMessages(library(NMF))
load("DrosophilaExpressions.rda")
set.seed(42)
```

**Original observations: first 16 genes**

```
imageBatchDisplay(V[, 1:16])
```



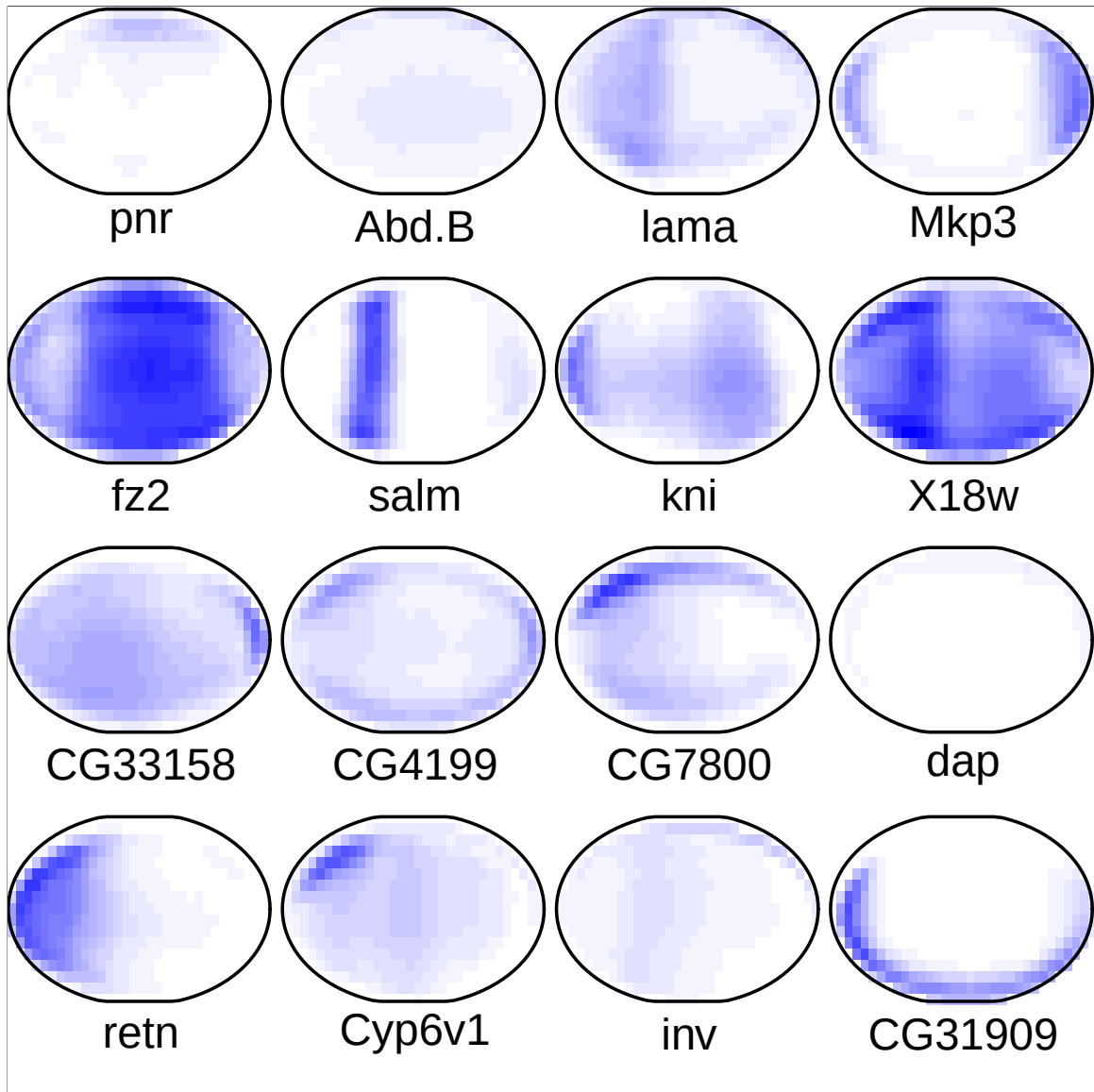
### Rank 15 factorization

```
res15 <- nmf(V, rank = 15, seed = 123, method = "brunet")
W15 <- basis(res15) # could also use res15@fit@W
H15 <- coef(res15) # could also use res15@fit@H
Vhat15 <- W15 %*% H15 # could also use fitted(res15)
cat("Generalized KL divergence (rank 15):", residuals(res15), "\n")
```

Generalized KL divergence (rank 15): 3626.139

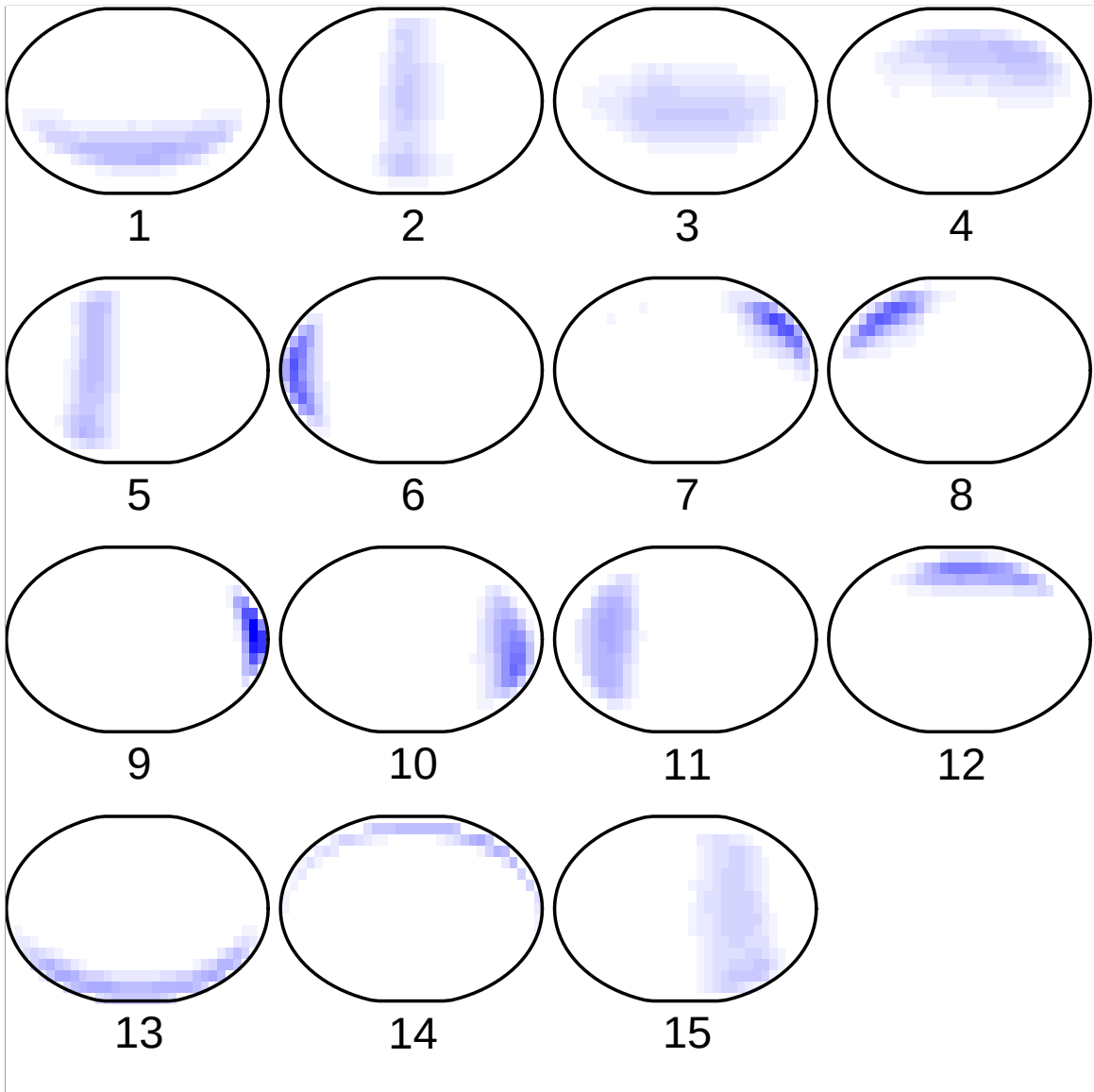
Approximated patterns of the first 16 genes ( $\hat{V}[1:16]$ ):

```
imageBatchDisplay(Vhat15[, 1:16])
```



The 15 basis patterns (columns of  $W$ ):

```
imageBatchDisplay(W15)
```



**Rank 10 factorization**

```

res10 <- nmf(V, rank = 10, seed = 123, method = "brunet")
W10 <- basis(res10)
H10 <- coef(res10)
Vhat10 <- W10 %*% H10
cat("Generalized KL divergence (rank 10):", residuals(res10), "\n")

```

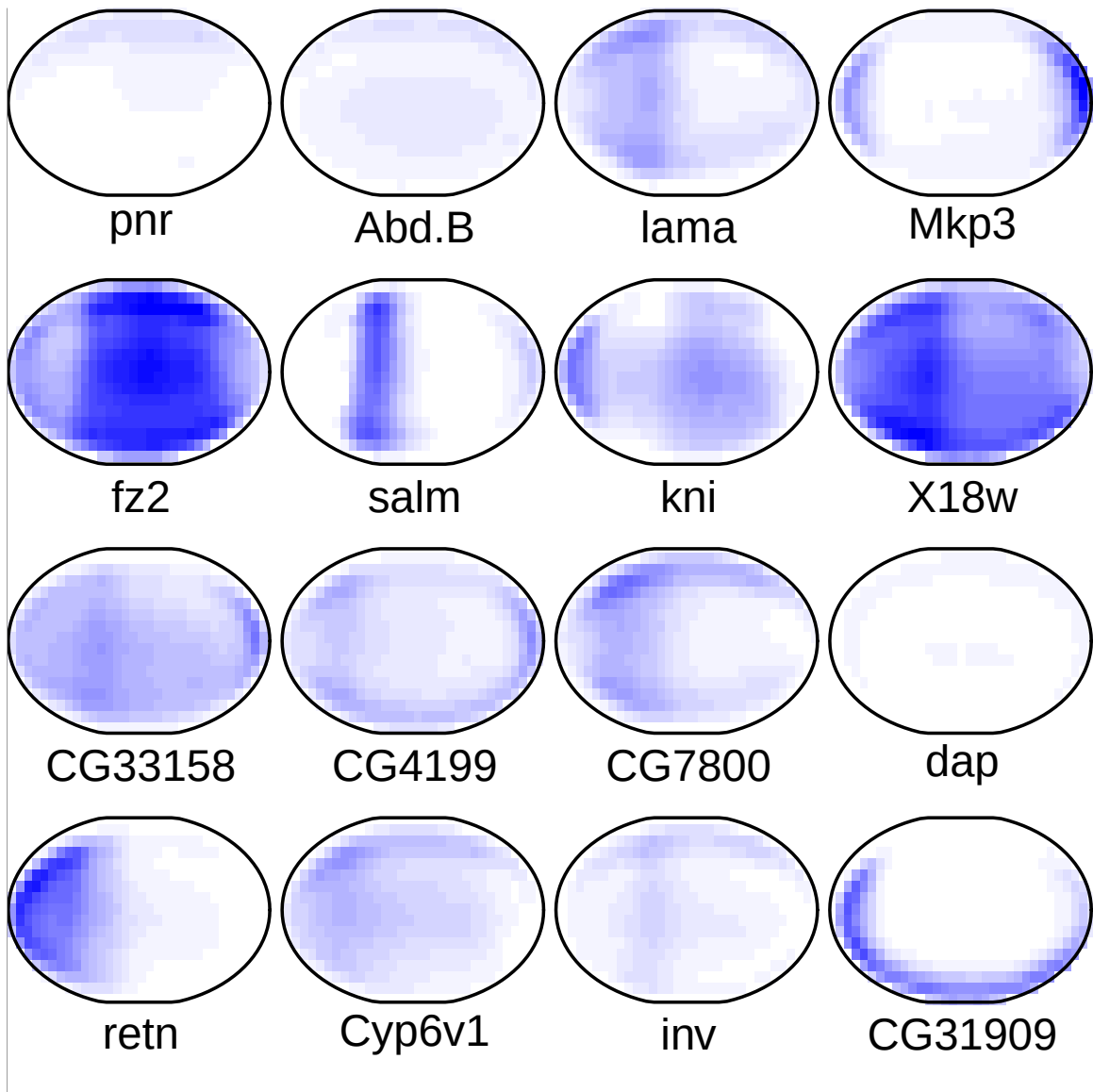
Generalized KL divergence (rank 10): 4901.298

Approximated patterns of the first 16 genes ( $\hat{V}[1:16]$ ):

```

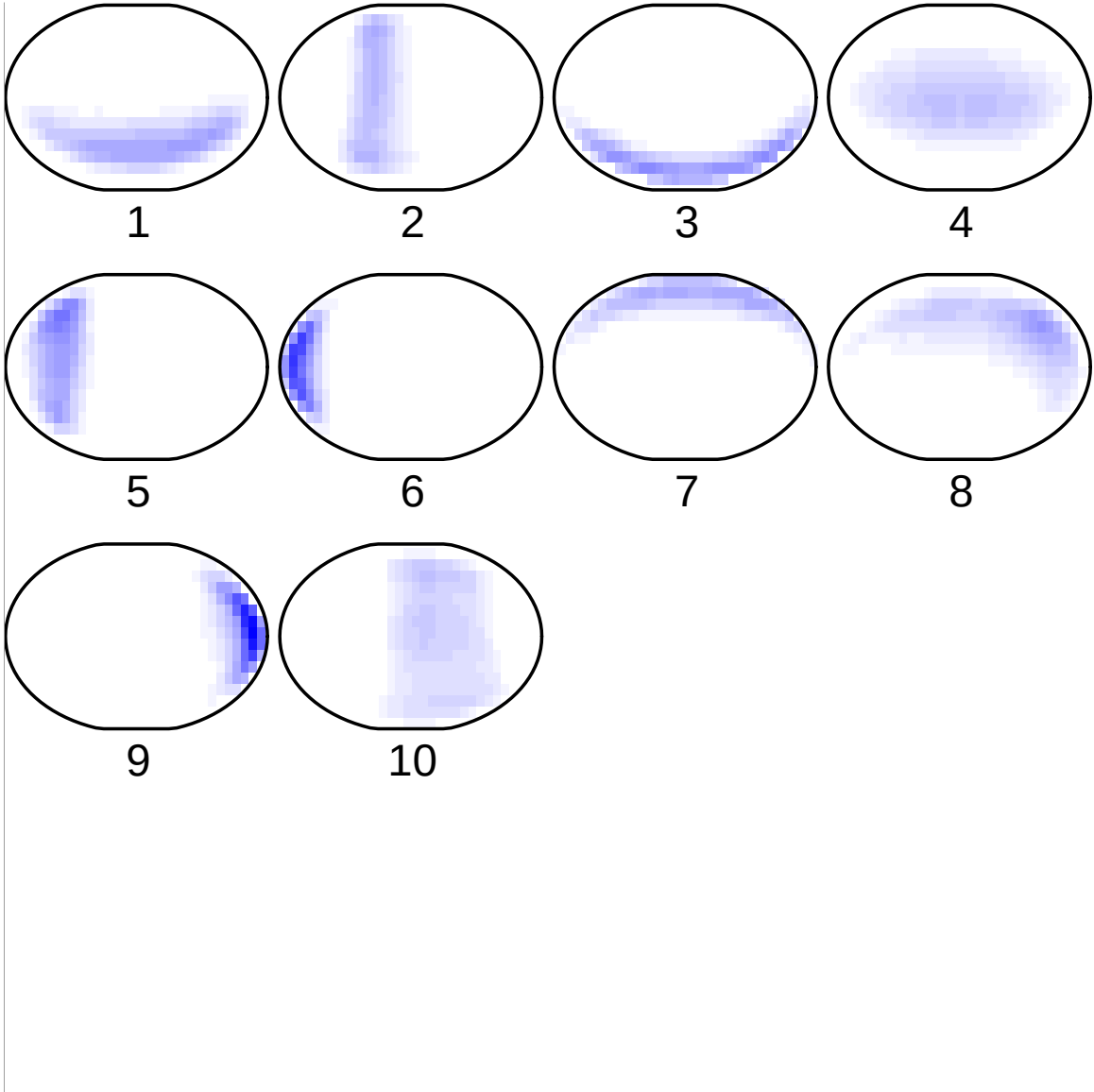
imageBatchDisplay(Vhat10[, 1:16])

```



The 10 basis patterns (columns of  $W$ ):

```
imageBatchDisplay(W10)
```



The KL divergence is lower for rank 15 ( $\approx 3626$ ) than for rank 10 ( $\approx 4901$ ), which is expected. A higher rank gives a richer basis and therefore a closer approximation. The rank-10 basis patterns are visibly broader / less specialized than the rank-15 ones.

## Part 2: Mkp3 and CG31909 as linear combinations of basis patterns

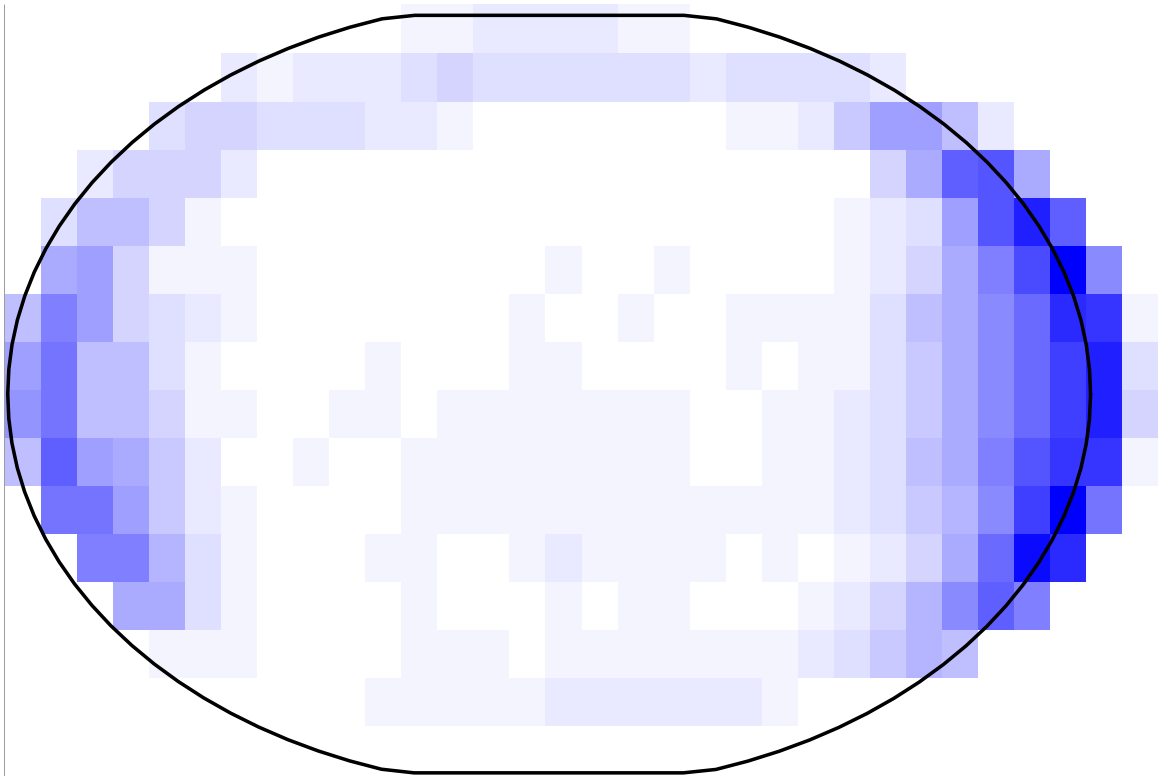
For a gene  $j$ , the approximated expression pattern is

$$\hat{V}_{\cdot j} = \sum_{k=1}^K H_{kj} W_{\cdot k},$$

i.e. a non-negative linear combination of the basis patterns  $W_{\cdot k}$  with weights  $H_{kj}$ .

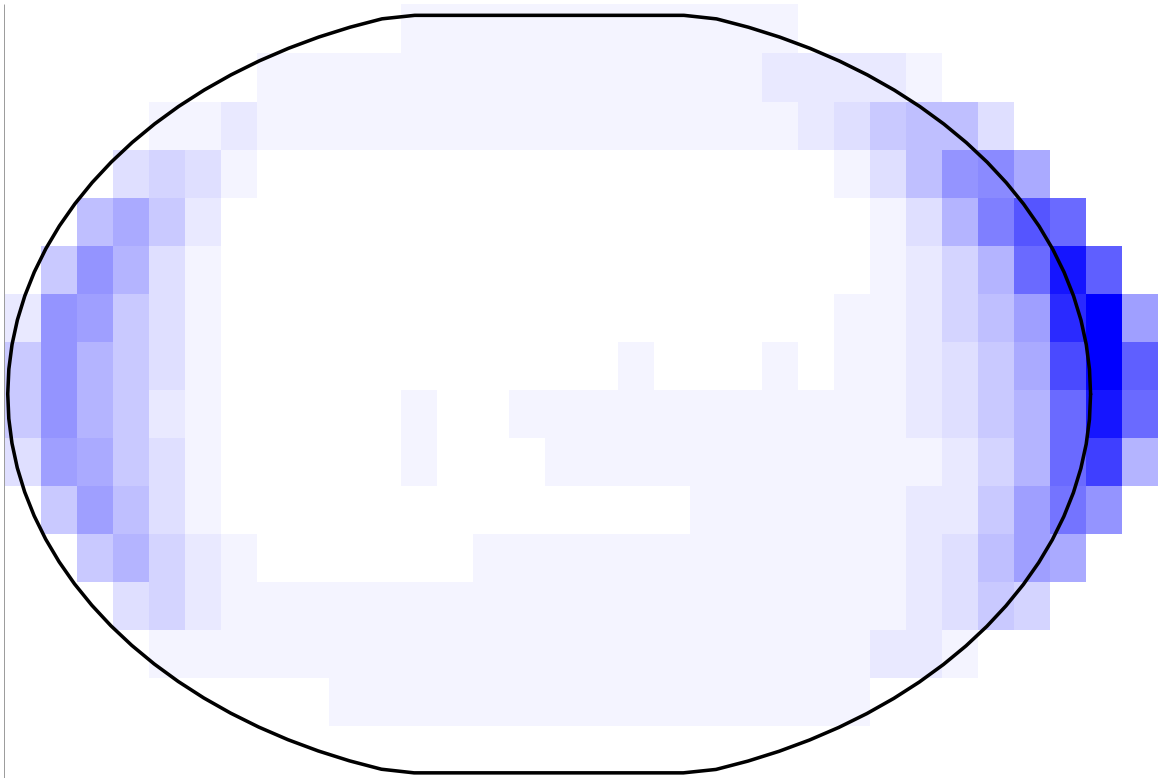
### Mkp3

```
mkp3_idx <- which(colnames(V) == "Mkp3")  
  
# True expression pattern of Mkp3  
imageBatchDisplay(V[, mkp3_idx, drop = FALSE], imgNames = "Mkp3 (observed)")
```



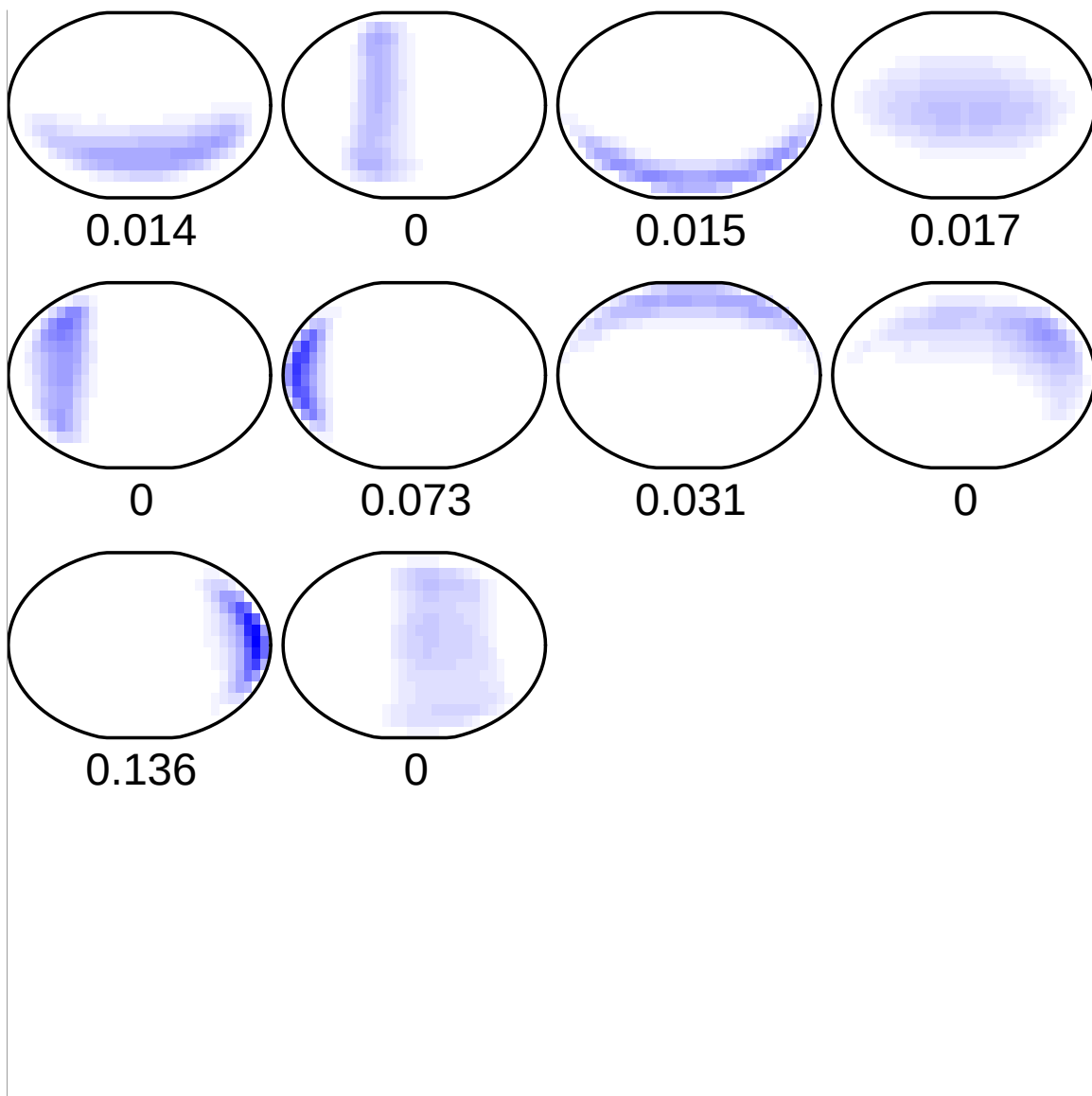
Mkp3 (observed)

```
# Reconstructed pattern of Mkp3
imageBatchDisplay(Vhat10[, mkp3_idx, drop = FALSE],
                  imgNames = "Mkp3 (reconstructed)")
```



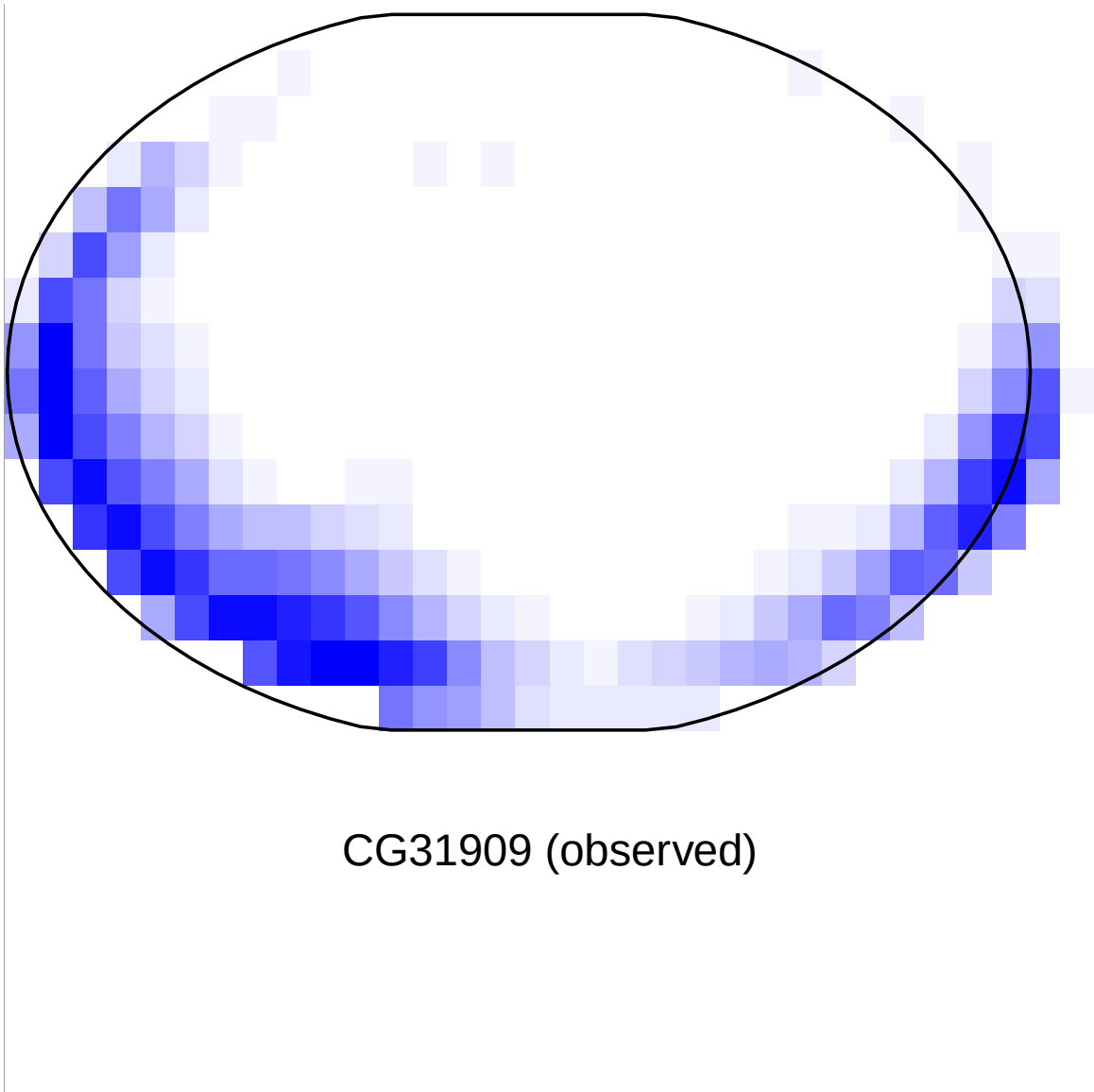
Mkp3 (reconstructed)

```
# Basis patterns labelled with the coefficients H[, Mkp3]  
imageBatchDisplay(W10, imgNames = round(H10[, mkp3_idx], 3))
```

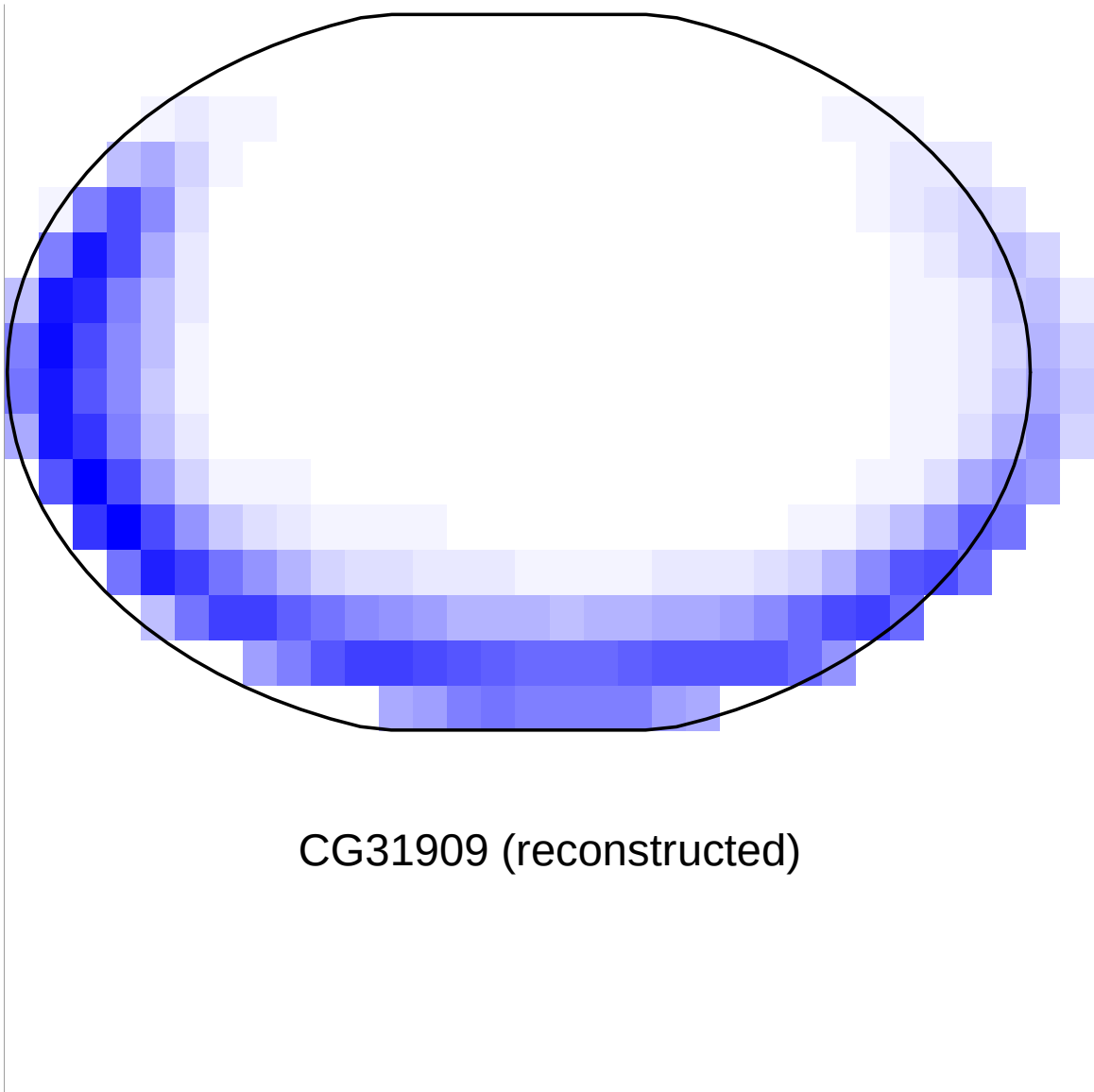


### CG31909

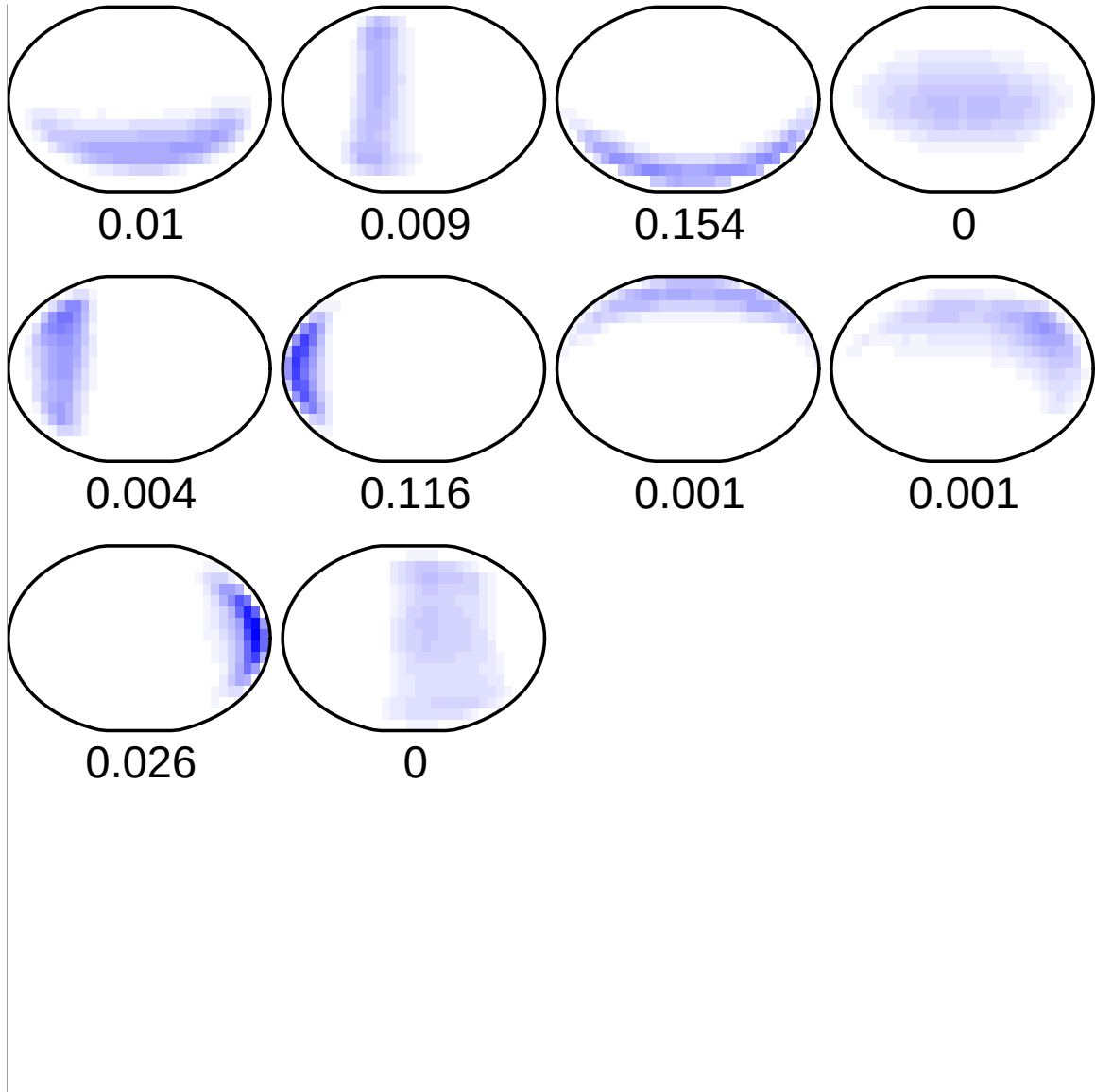
```
cg_idx <- which(colnames(V) == "CG31909")
imageBatchDisplay(V[, cg_idx, drop = FALSE], imgNames = "CG31909 (observed)")
```



```
imageBatchDisplay(Vhat10[, cg_idx, drop = FALSE],  
                  imgNames = "CG31909 (reconstructed)")
```



```
imageBatchDisplay(W10, imgNames = round(H10[, cg_idx], 3))
```



### Interpretation

The 10 columns of  $W$  form a parts-based dictionary of canonical spatial expression motifs of the early *Drosophila* embryo: anterior caps, posterior caps, lateral stripes, dorsal and ventral domains, terminal regions, etc. They are not raw observations, but recurring spatial templates that the algorithm extracts because most gene patterns can be reconstructed as a sum of just a few of them.

The coefficients  $H_{kj}$  are the loadings of gene  $j$  on each motif:  $H_{kj}$  tells us how strongly motif

$k$  contributes to gene  $j$ 's spatial expression, with  $H_{kj} = 0$  meaning that motif is irrelevant for that gene. Because both  $W$  and  $H$  are non-negative, the contributions only add up and components cannot cancel each other out, which is what makes the decomposition naturally interpretable in terms of which spatial regions does this gene express in.

For Mkp3 and CG31909, only a small number of basis patterns receive a substantially non-zero coefficient, which means each gene is approximated as a sparse mixture of a few canonical regions. The reconstructed images closely resemble the observed patterns, confirming that these few basis components capture the dominant spatial structure of the gene.

## Problem 26: Implementing NMF from scratch

The key challenge of this problem is to code the update rules correctly. The authors try to explain this below.

Let's consider the multiplicative update formula for  $H$ :

$$H_{kj} \leftarrow H_{kj} \frac{\sum_i W_{ik} V_{ij} / (WH)_{ij}}{\sum_i W_{ik}}$$

Notice that the numerator in the fraction can be rewritten as:

$$U_{kj}, U := (W^T (V / (WH)))$$

where the  $/$  denotes elementwise division of two matrices.

The denominator in the fraction in the formula requires that every row  $k$  of  $U$  needs to be divided by the sum of the entries in column  $k$  of  $W$ . These are the observations that lead to the implementation of the update step for  $H$  in the code below.

Let's shift now our attention to the multiplicative update for  $W$ :

$$W_{ik} \leftarrow W_{ik} \frac{\sum_j H_{kj} V_{ij} / (WH)_{ij}}{\sum_j H_{kj}}$$

Notice that the numerator in the fraction can be here rewritten as:

$$Q_{ik}, Q := ((V / (WH)) H^T)$$

The denominator in the fraction of this formula requires that every column  $k$  of  $Q$  needs to be divided by the sum of the entries in row  $k$  of  $H$ . These operations are also implemented in the code below.

Now, we apply the 1000 iterations where we perform a vectorized version of the multiplicative update at each step.

```
load("DrosophilaExpressions.rda")

# Generalized KL divergence with log 0 = 0
gkl <- function(V, V_hat) {
  V_pos <- V > 0 # V_pos are the indices of non zero numbers
  # which are the only ones considered in the next line, so that 0*log(0) = 0
  sum(V[V_pos] * log(V[V_pos] / V_hat[V_pos])) - sum(V) + sum(V_hat)
}

nmf_kl <- function(V, K, n_iter = 1000, seed = 123) {
  set.seed(seed)
  M <- nrow(V); N <- ncol(V)
  W <- matrix(runif(M * K), M, K)
  H <- matrix(runif(K * N), K, N)
  eps <- .Machine$double.eps # to avoid that division by 0 gives error, a very small number

  for (it in seq_len(n_iter)) {
    # H update: H is K x N
    # Dividing recycles along rows of H (each row k divided by colSums(W)[k]).
    WH <- W %*% H + eps
    num_H <- t(W)%*%(V/WH)
    H <- H * (num_H / colSums(W))

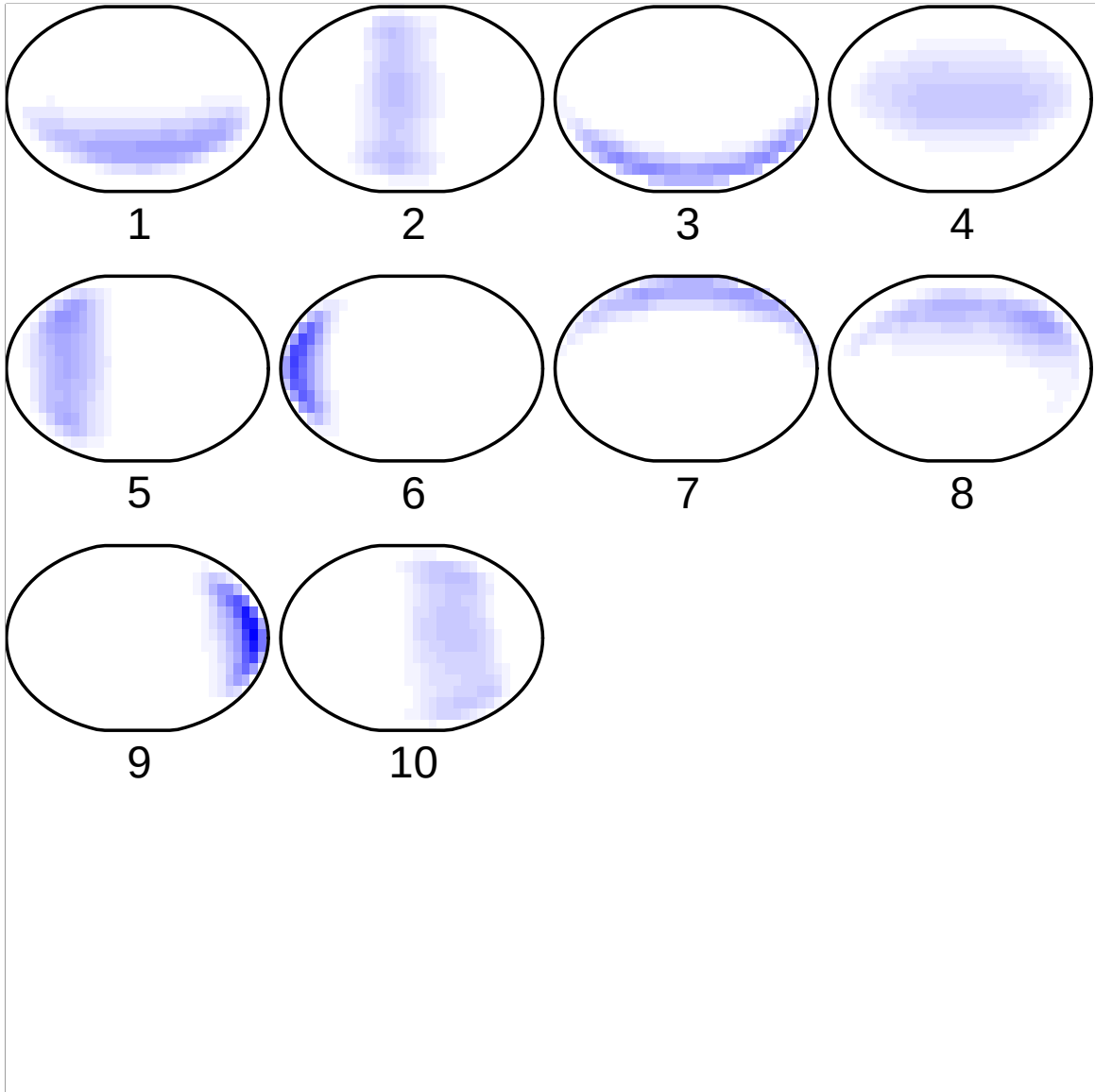
    # W update: W is M x K
    # We need to divide column k of W by rowSums(H)[k], which is column-wise scaling.
    WH <- W %*% H + eps
    num_W <- (V/WH)%*%t(H)
    W <- W * t(t(num_W) / rowSums(H))
  }
  list(W = W, H = H) # function will output the basis W and coeffs H
}

fit <- nmf_kl(V, K = 10, n_iter = 1000, seed = 123)
W_self <- fit$W
H_self <- fit$H
cat("Generalized KL divergence after 1000 iters:",
    gkl(V, W_self %*% H_self), "\n")
```

Generalized KL divergence after 1000 iters: 4960.585

Inferred basis patterns:

```
imageBatchDisplay(W_self)
```



The KL divergence reached by our hand-written algorithm (4960) is close to the value reported by the **NMF** package for the same rank (4901). The visualized basis patterns also closely resemble the ones obtained from the **NMF** package. They capture the same canonical spatial regions of

the embryo (anterior, posterior, etc.), differing only by the order of the components and small differences in detail. This is consistent with the non-uniqueness of NMF discussed in Problem 24. The two algorithms can reach different but essentially equivalent factorizations differing by a permutation and rescaling of the components, plus minor numerical differences from random initialisation.